Fundamental Neuroscience for Neuroimaging — Week 1

The Anatomy Behind the Image

Structure, Development and Blood Supply of the Human Brain

Four modules of groundwork: what neuroscience and neuroimaging are and why the data have become a big-data problem; the cells the brain is built from and the ways anatomists carve it up; how brains compare across species and how they develop; the blood supply, which matters more than it first appears because fMRI measures blood rather than neurons; the terminology used to state where anything is; how neurons signal to each other and to the body; the functional systems whose jobs are well established; what damaged brains revealed about language, personality and memory; how cognition is measured systematically; and the imaging methods themselves — what each technique measures, where the MRI and fMRI signals come from, how studies using them are designed, and what the brain is doing when you ask it to do nothing. Green boxes give the plain-words version of each idea; rust boxes flag places where terminology or a stated fact needs care.

Parts are named by content rather than by module number: the source decks label themselves inconsistently, with two different decks both called "Module 3" and the vascular material appearing under two different numbers.

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Part I

Introduction

What neuroscience is, what neuroimaging is, and why the data are large.

1. Why This Course Exists

[Lecture slide omitted from the public copy.] Publications using MRI methods, and the first MRI scanner with the first scan of a torso.

The idea in plain words

The method is young and the growth has been steep. A handful of papers in the early 1990s became over **16,000 publications** using magnetic resonance imaging in 2014 alone.
The data have grown even faster than the literature. The first study, in 1977, used **106 data points**. Recent single studies report over **10 billion**.

The idea in plain words

Multi-site consortia push this further still. The **Alzheimer's Disease Neuroimaging Initiative** (ADNI) covers thousands of subjects, at multiple sites, each scanned repeatedly over years — so the dataset has structure in space, in time, and across scanners.
That is why the lecture calls modern neuroimaging analysis a genuine **big data** problem: not merely large, but heterogeneous in ways that the analysis has to model rather than ignore.

2. Neuroscience and Neuroimaging

[Lecture slide omitted from the public copy.] Neuroscience and its constituent disciplines.

Neuroscience

The **multidisciplinary study of the biological basis of behaviour**.

The idea in plain words

It is worth separating two lists that are easy to conflate.
Disciplines — distinguished by *what they measure*:

- **neuroanatomy** — the structures of the brain
- **neurochemistry** — the chemistry of communication between cells
- **neurophysiology** — the electrical properties of the brain
- **neuropsychology** — cognitive domains and the structures supporting them

Branches — distinguished by *what question is being asked*: molecular, cognitive, clinical, computational, developmental and cultural neuroscience, among others.

Neuroimaging

A **collection of methods** to image the structural, functional and chemical properties of the central nervous system.

Watch out — neuroimaging is a method, not a field

This distinction runs through the whole course. Neuroimaging is not a branch of neuroscience sitting alongside the others — it is a set of tools *used by* many disciplines and many branches. The practical consequence is that a scan means nothing on its own. Interpreting it requires whichever body of anatomy, physiology or psychology the question belongs to, which is exactly why a course on imaging opens with four modules of anatomy.

[Lecture slide omitted from the public copy.] What the course will cover.

The idea in plain words

The topics: the structural and functional organisation of the brain; the terminology of brain organisation; brain networks and communication; cognition and cognitive domains; the principles of magnetic resonance imaging; neuroimaging methods; and experimental design. Note the ordering. Anatomy and terminology come first, MRI physics only later — the course treats knowing what you are looking at as prior to knowing how the picture was made.

Part II

Structural Anatomy

The two cell types, the parts of a neuron, and four ways of dividing up the brain.

3. Two Kinds of Cell

[Lecture slide omitted from the public copy.] Neurons and glia.

The brain's two cell categories

Neurons — highly specialised, interconnected cellular units; the fundamental *processing* unit of the brain and of the central nervous system.
Glia — the non-neuronal *supportive* cells, forming the brain's scaffolding.

[Lecture slide omitted from the public copy.] One neuron in detail, and the variety of cell types found in different brain locations.

The idea in plain words

Neurons come in many types, shapes, sizes and functions, and which type you find depends on where in the brain you look. The word "neuron" names a role rather than a single design.

4. The Parts of a Neuron

[Lecture slide omitted from the public copy.] The four basic elements.

Axon

Varies in length and complexity — from a few millimetres to spanning one side of the brain to the other. It **transmits** the electrical signal away from the cell body, towards target neurons.

Cell body (soma)

Contains the **nucleus**, and therefore the DNA. Also holds the specialised organelles — **mitochondria**, **Golgi apparatus**, **ribosomes** and polysomes — that supply energy and synthesise the proteins needed to generate and propagate the signal along the axon.

Dendrites

Extend from the cell body and form the **receiving** contacts from other neurons. They branch in complex patterns and carry **dendritic spines**, small protrusions that change with experience and form the physical basis of learning.

Axon terminals

Also called **presynaptic terminals**. Very fine branches that form the communication sites with other neurons — most ending on the next neuron's dendrites, some on its cell body.

[Lecture slide omitted from the public copy.] Dendrites from the hippocampus, a structure critical for memory.

The idea in plain words

The four parts describe one direction of flow: *dendrites receive*, the *cell body integrates and powers*, the *axon transmits*, and the *terminals hand over* to the next cell.
Two details are worth holding onto because later modules build on them. Dendritic spines change with learning — structure is not fixed. And axon length varies enormously, which is

what makes long-range communication between distant brain regions physically possible.

5. Glia

[Lecture slide omitted from the public copy.] The supporting cells.

The idea in plain words

Glia *outnumber* neurons in the brain, which is easy to forget given how much attention neurons receive. Their roles:

- forming connective tissue and the structure of the brain
- metabolic support for neurons
- removing excess neuronal secretions
- producing **myelin**, which insulates axons and speeds electrical propagation along them

Watch out — myelin is why "white matter" is white

This one term connects the cell biology to what you will actually see in a scan. Myelin is fatty, and fat appears bright on a T1-weighted MRI.
White matter is tissue dominated by myelinated axons — the wiring. **Grey matter** is dominated by cell bodies and dendrites — the processing. So the light/dark contrast on a structural scan is, at root, a map of where the glia have laid down myelin.

6. Four Ways of Dividing the Brain

The idea in plain words

Large groups of similar, spatially organised neurons form the brain's structures and networks. The lecture gives four different schemes for carving the brain up — and they are complementary rather than competing, each answering a different question.

1. Cytoarchitectural organisation

[Lecture slide omitted from the public copy.] Brodmann's cytoarchitectural maps, for humans and for other species.

Cytoarchitecture

Structural organisation based on **cellular composition** — which cell types are present and how they are layered.
Mapped by the German anatomist **Brodmann**, who published his cytoarchitectural map of the human brain in **1909**. **Brodmann areas** are still routinely used to report where a neuroimaging finding is located.

The idea in plain words

A century-old map made by staining dead tissue and looking down a microscope remains standard vocabulary in papers reporting results from scanners that did not exist when it was made. That persistence is itself informative: the boundaries Brodmann drew from cell composition turned out to correspond to real functional divisions.
Brodmann also mapped the hedgehog, rabbit, marmoset and lemur, which allowed cell types and their locations to be compared across species — a theme that returns in Module 4.

2. Dissociable brain structures

[Lecture slide omitted from the public copy.] Printed and electronic brain atlases.

The idea in plain words

Many structures can be identified by cell type and anatomical boundaries, either on an MRI scan or in post-mortem tissue. This has produced highly detailed **brain atlases**, both printed and as searchable electronic resources giving each structure's location and proposed function.

3. Dissociable brain networks

[Lecture slide omitted from the public copy.] Structures grouped into networks by their connections.

Brain networks

Groups of structures that form a functional unit by virtue of the **connections between them**. Two examples:
Striatum — nucleus accumbens, caudate nucleus and putamen: highly interconnected, forming a single functional network.
Medial temporal lobe — hippocampus, entorhinal and perirhinal cortex: critical for memory formation.

The idea in plain words

This is a different *kind* of division from the first two. Cytoarchitecture and atlases divide by what tissue *is*; networks divide by what it is *connected to*.
A network need not be one contiguous lump, and that is precisely the point — it is defined by wiring rather than by adjacency.

4. The six lobes

[Lecture slide omitted from the public copy.] The rudimentary anatomical classification.

The six lobes

Frontal, parietal, occipital, temporal, limbic, and the insular cortex.

Watch out — six, not four

The familiar textbook figure names four lobes — frontal, parietal, occipital, temporal. This scheme adds the **limbic lobe** and the **insular cortex**, both of which are less visible from the outside: the insula sits buried beneath the frontal and temporal lobes, and the limbic lobe forms a ring on the brain's inner surface.

The idea in plain words

The lecture calls this classification "rudimentary" and then defends it: it remains valid because those broad structural areas turn out to share common *functions*.
That is the recurring justification behind all four schemes. Each survives because a division drawn on anatomical grounds — cells, boundaries, connections, gross shape — turns out to predict something functional.

Part III

Developmental Organisation

How brains compare across species, and the order in which they are built.

7. The Same Brain, Different Sizes

[Lecture slide omitted from the public copy.] Great variability in size and shape; great overlap in organisation.

The idea in plain words

Compared across species — from a frog up to a human — brains vary enormously in size and shape. But cross-sections reveal substantial *overlap in basic organisation*. The same parts are present; they differ in proportion.

[Lecture slide omitted from the public copy.] Human and monkey, with the shared structures labelled.

Structures shared across species

Cortex — the outer sheet; differs most in size.
Brainstem (with the spinal cord) — very similar across species.
Thalamus — a relay station, important for communication within the brain.
Cerebellum — critical for fine motor control.

[Lecture slide omitted from the public copy.] Human and chimpanzee: corpus callosum, hippocampus, thalamus and midbrain.

Two more

Corpus callosum — a bundle of axons carrying communication between the two hemispheres.

Hippocampus — critical for memory formation; very similar in organisation and structure between animals and humans.

[Lecture slide omitted from the public copy.] Human, cat and mouse: the subcortical structures are alike; the cortex is not.

The idea in plain words

The mouse brain has the same basic parts — ventricles, striatum, thalamus. Comparing mouse, cat and human, the **largest difference by far is the size of the cortex**, while the subcortical structures stay much the same.
This is the practical payoff of the whole module: because the shared structures really are shared, experiments in simpler animals can inform human neuroscience.

Watch out — the shared parts are the ones you can generalise from

The argument cuts both ways, and the asymmetry is worth stating plainly. Work on hippocampal memory circuits or brainstem function in a mouse transfers to humans relatively well, because those structures are conserved.
Work that depends on the cortex — which is exactly where humans differ most — transfers least well. The same figure that justifies animal models also marks their limit.

8. How the Brain Is Built

[Lecture slide omitted from the public copy.] From the embryonic neural tube to the newborn brain.

The idea in plain words

Development proceeds from the embryonic **neural tube** in a fixed order:
1. the **spinal cord**
2. the **hindbrain** — including the cerebellum
3. the **midbrain** — the tectum and tegmentum
4. the **forebrain** — including the cortex

The idea in plain words

The ordering carries the module's point. The **cortex forms last** — and the cortex is also what differs most between humans and simpler animals.
Development recapitulates the cross-species comparison: the structures built earliest are the ones most conserved across species, and the structure built last is the one that expanded most in us.

Watch out — a slip in the narration

The lecture describes the thalamus and midbrain as "important for basic life functions, breathing, heart rate". Breathing and heart rate are controlled by the **brainstem** — specifically the medulla and pons — rather than by the thalamus, whose role, as the same lecture says elsewhere, is that of a relay station.

The broader claim being made is sound: the deep, evolutionarily older structures handle the basic life-sustaining functions. It is the attribution to the thalamus specifically that does not hold.

Part IV

Vascular Anatomy

The blood supply, and why an imaging course cares about it.

9. Why Blood Matters Here

[Lecture slide omitted from the public copy.] An angiogram of the blood supply to the head, and the density of vessels at increasing depth.

The idea in plain words

The brain consumes a tremendous amount of energy and oxygen, supplied by an intricate system of arteries and veins. In the images, the cortical surface is at the top; moving deeper reveals progressively finer vessels supplying the neurons and glia in each region.

Watch out — this module is really about fMRI

Anatomy of blood vessels can look like a detour in an imaging course. It is not. **fMRI does not measure neural activity** — it measures a blood-related signal, and infers activity from it.
Everything in this module therefore constrains what an fMRI result can mean: where blood can be delivered, how quickly, and at what spatial grain. The lecture flags this explicitly, saying the vascular territories matter "for neuroimaging methods that we will discuss down the line".

10. The Route In

[Lecture slide omitted from the public copy.] From the aorta to the Circle of Willis.

The idea in plain words

The path, in order:

1. the **aorta**, leaving the heart
2. the **common carotid artery**, which splits into two
3. the **external carotid artery** — supplying the face and skull
4. the **internal carotid artery** — the main blood supply to the brain
5. the **Circle of Willis**, where the blood enters the brain

The external/internal split is the one to remember: only one branch of that fork is carrying blood to the brain at all.

[Lecture slide omitted from the public copy.] The Circle of Willis at the base of the brain, with its branches labelled, and a schematic of the same vessels.

The Circle of Willis

A ring of connected arteries at the base of the brain, from which blood is distributed to specific brain areas.

The idea in plain words

A *ring* is an unusual arrangement for plumbing, and the shape is the point. Because the vessels form a closed loop, blood arriving from one side can reach territories normally fed from the other.
It is a redundancy: if one supplying artery narrows or is blocked, the circle offers an alternative route. That collateral capacity varies a great deal between individuals.

[Lecture slide omitted from the public copy.] The major arteries listed in the lecture.

The vessels named

Anterior cerebral artery; middle cerebral artery; posterior cerebral artery; superior cerebellar artery; pontine arteries; anterior inferior cerebellar artery; vertebral artery; posterior inferior cerebellar artery.

Watch out — the list mixes vessels that *feed* the circle with vessels that *leave* it

The slide heads this list "Circle of Willis", which reads as though all eight branch off it. Two do not, and the distinction is worth having straight:

- **Anterior circulation** — the **internal carotid** arteries feed the circle, and the anterior and middle cerebral arteries branch from it.
- **Posterior circulation** — the **vertebral** arteries join to form the **basilar** artery, which feeds the back of the circle. The posterior inferior cerebellar artery branches from the vertebral artery, *before* the circle.

So the vertebral artery and PICA belong to the system supplying the circle rather than draining from it.
Relatedly, the Module 4 narration states that "the middle cerebral artery is the primary blood supply to the circle of Willis". That inverts the relationship — the MCA is one of the largest arteries leaving the circle, and the internal carotid is what supplies it. Module 3 states this correctly, so the two modules disagree; Module 3 is the one to follow.

11. Vascular Territories

[Lecture slide omitted from the public copy.] Cortical vascular territories: which artery supplies which part of the cortex.

The idea in plain words

Because the supply is so organised, the cortex can be divided into **vascular territories** — regions each fed by one particular artery. The boundaries are fairly specific, with only a little overlap.

On the lateral surface, the **middle cerebral artery** covers a large central swathe, the **anterior cerebral artery** the upper and frontal margin, and the **posterior cerebral artery** the occipital region at the back.

Watch out — territories are why strokes produce characteristic deficits

If each artery supplies a delineated region, then a blockage in a named artery damages a predictable set of structures — and therefore causes a predictable pattern of deficits.
This is what makes the map clinically useful rather than merely tidy, and it is the reason vascular territories are worth memorising in a way that most plumbing is not.

The idea in plain words

Note the mild tension between this and the previous section. The Circle of Willis provides redundancy; vascular territories are nonetheless fairly sharply delineated. Both are true: the circle offers a fallback route *into* the system, while beyond it each artery still serves its own region with little overlap.

12. Where This Leaves Us

- **Neuroimaging is a method used across neuroscience**, not a branch of it — which is why interpreting a scan needs anatomy first.
- **The brain is built from neurons and glia**, with glia the more numerous, and their myelin is what gives white matter its appearance on a structural scan.
- **Four schemes divide the brain** — by cell composition (Brodmann), by anatomical boundaries (atlases), by connectivity (networks), and by gross shape (six lobes). Each survives because it predicts something functional.
- **Brains differ across species mostly in the cortex**, which is also the structure that develops last — so animal models transfer best for the conserved subcortical structures and worst for what makes us distinctive.
- **Blood reaches the brain through the internal carotid and vertebral arteries** and is distributed from the Circle of Willis, whose ring shape provides collateral supply.
- **Each artery serves a delineated cortical territory**, which explains stroke syndromes — and, because fMRI measures a blood-derived signal, sets a limit on what functional imaging can resolve.

Part V

The Terminology of Brain Organisation

The axes, the planes, and the left–right problem that the images themselves cannot solve.

13. Why a Standard Vocabulary

The idea in plain words

A standard nomenclature is used across vertebrate nervous systems, and particularly in MRI research, to describe locations and physical relationships in the brain.
It is the language in which results get reported — both where a structure sits and where an activation was found in an fMRI study. None of it is conceptually difficult; it simply has to be learnt.

14. Axes: Animals and Humans

[Lecture slide omitted from the public copy.] The two axes as defined for a four-legged animal.

The animal axes

Rostral–caudal — the major (long) axis. *Rostral* from *rostrum*, a beak: the front. *Caudal* from *cauda*, a tail: the back.
Dorso–ventral — the vertical axis. *Dorsal* is the back (as in the dorsal fin); *ventral* is the belly.

Watch out — standing upright breaks the scheme

For a four-legged animal the two axes are perpendicular and unambiguous. Humans are upright and the brain is tilted relative to the body axis, so the animal terms stop lining up. Applied literally to a human: caudal is still the back of the head and dorsal still the top, rostral is still the front — but *ventral* would also point to the front, since that is where the belly is. Two different terms end up meaning the same direction, which is exactly why a separate human vocabulary exists.

[Lecture slide omitted from the public copy.] The human terms.

The human terms

Anterior — front. **Posterior** — back.
Superior — top. **Inferior** — bottom.
Medial — towards the centre axis of the body or brain. **Lateral** — away from it.

The idea in plain words

In practice both vocabularies appear in the literature, and the animal terms have not gone away — "dorsolateral prefrontal cortex" and "ventral striatum" are standard names. Reading anatomy papers means holding both sets at once.

15. Planes

[Lecture slide omitted from the public copy.] The three planes of orientation.

[Lecture slide omitted from the public copy.] The same three planes, under the names used for MRI images.

Three planes, two sets of names

Anatomical	MRI	Cuts the brain into
frontal	**coronal**	front and back
transverse (horizontal)	**axial**	top and bottom
sagittal	**sagittal**	left and right

The sagittal plane runs from the back of the head through the nose; the midline version is the **sagittal midline**.

The idea in plain words

Only the first two are renamed for imaging. *Sagittal* keeps its name in both vocabularies, so there are three planes but only five words to remember.

16. Describing Relative Position

[Lecture slide omitted from the public copy.] A sagittal view: the cerebellum is posterior and *ventral to the frontal cortex.*

[Lecture slide omitted from the public copy.] A coronal slice: the hippocampus is inferior and *lateral to the third ventricle.*

The idea in plain words

The terms are used in combination, because one axis rarely locates anything on its own. Two structures can share a superior–inferior level and still be far apart medially, so a usable description names a position on more than one axis at once.
These two examples are worth reading off the images directly — that is the skill the module is teaching, rather than the definitions themselves.

17. Left Versus Right

[Lecture slide omitted from the public copy.] A coronal MRI — and nothing in it says which side is which.

Watch out — the image is genuinely ambiguous

Look at a coronal slice and ask which side is the patient's left. The image cannot tell you. The identical picture would be produced whether you stood *behind* the person looking forward, or *in front* of them looking back.
The brain is roughly symmetric, so there is no internal landmark to settle it. This is not a subtlety — it is a genuine hole that has to be plugged by convention or by a physical marker.

[Lecture slide omitted from the public copy.] The two conventions.

Radiological and neurological convention

Radiological convention — the patient's **left** appears on the **right** of the image. As though you were facing the person.
Neurological convention — the patient's **left** appears on the **left** of the image. As though you were standing behind them.

The idea in plain words

A mnemonic that avoids memorising the pair: *neurological* is the "naive" one — left is left, right is right. Radiological is the flipped one, and it is flipped because a radiologist is conventionally imagined facing the patient, as in a physical examination.

[Lecture slide omitted from the public copy.] Both conventions are in use, and which one applies is usually set by the analysis software.

Watch out — this is a real source of published error

Both conventions are used in research, and which one you get is usually decided by the analysis software rather than by the researcher. Consistency therefore has to be maintained deliberately.
The failure mode is specific and serious: if images are stored in radiological convention but read as neurological, an fMRI finding gets reported on the *wrong hemisphere*. Nothing in the image looks wrong — the analysis runs normally, the statistics are valid, and the anatomical label is simply mirrored.

Fiducial marker

A magnetic or metallic object — sometimes just a vitamin pill — placed in the field of view on a known side (conventionally the right) before acquisition.
It appears in the image, so which side is which becomes visible rather than assumed.

The idea in plain words

The fiducial marker is the honest engineering answer to the problem. Rather than trusting a software setting and a chain of assumptions, put a physical object in the scanner and read the answer off the picture.

18. A Few More Terms

[Lecture slide omitted from the public copy.] The remaining vocabulary.

Also commonly used

Proximal — closer to the point of attachment. **Distal** — further from it.
Ipsilateral — the same side of the brain. **Contralateral** — the other side.
Oblique plane — a plane at an angle, neither horizontal nor vertical.

The idea in plain words

Contralateral is worth flagging as the one that recurs most. Much of the nervous system is crossed — the left hemisphere controls the right hand, and visual input from the right field arrives in the left hemisphere — so "contralateral" appears constantly in results, and misreading it inverts the claim being made.

19. Where This Leaves Us

- **Animal and human axis terms differ** because standing upright makes the animal scheme ambiguous — but both vocabularies survive in the literature and in structure names.
- **Three planes, five names.** Frontal/coronal, transverse/axial, and sagittal, which keeps its name in both.
- **Positions are stated on several axes at once**, because one axis rarely locates a structure on its own.
- **An MRI image cannot show which side is left.** The same picture results from viewing the person from the front or from behind.
- **Two conventions resolve it** — radiological (left appears right) and neurological (left appears left) — and mixing them up reports a finding in the wrong hemisphere without anything looking amiss.
- **A fiducial marker settles it physically**, which is why studies use one rather than relying on a software setting.

Part VI

Communication in the Brain — the Synapse

How one neuron passes a signal to the next, and what an electrode actually records.

20. Two Historical Theories

Reticular theory versus the neuron doctrine

Reticular theory (Golgi) — neurons form a continuous net, physically joined to one another.
Neuron doctrine (Cajal) — neurons are *structurally independent* units that interact by **contiguity**, not **continuity**: they touch without merging.

The idea in plain words

Cajal reached his conclusion from meticulous staining and microscope work, and he was right — but the question was not settled until the **electron microscope** arrived in the 1950s and made the gap directly visible.
Everything downstream depends on this. If neurons were continuous, an electrical signal

could simply flow onwards. Because they are separated, the signal has to be converted into a chemical one and back again — which is what the rest of this part describes, and what gives pharmacology somewhere to intervene.

21. Three Types of Neuron

[Lecture slide omitted from the public copy.] Sensory, inter- and motor neurons.

The three types

Sensory neuron — converts an external stimulus into an electrical signal the brain can process.
Interneuron — processes and relays information once received.
Motor neuron — converts a processed electrical signal into muscle or gland action.

The idea in plain words

In one line: *in*, *process*, *out*. Neurons collectively receive and process information, transmit it both chemically and electrically, pass it downstream, and form the networks that support brain function.

22. The Resting Potential

[Lecture slide omitted from the public copy.] Ion channels and pumps in the membrane.

The idea in plain words

Ion channels and **ion pumps** in the membrane maintain different concentrations of **sodium**, **potassium**, **chloride** and **calcium** inside the cell than outside.
A difference in charge across the membrane is a voltage, and the cell's standing voltage is its **resting potential**. The neuron is therefore a charged battery that has not yet been discharged — which is what makes a fast signal possible at all.

23. Synaptic Transmission, Step by Step

[Lecture slide omitted from the public copy.] The synapse: presynaptic terminal, synaptic cleft, postsynaptic spine.

The sequence

1. An influx of **calcium** through an ion channel makes synaptic **vesicles** fuse with the presynaptic membrane.
2. **Transmitter** is released into the **synaptic cleft**.
3. Transmitter **binds** receptors on the postsynaptic membrane.
4. Postsynaptic channels **open or close** as a result.
5. Opening channels lets ions flow in; closing them prevents it.
6. The changed ion balance **depolarises** or **hyperpolarises** the postsynaptic cell.

7. If the voltage change is large enough, an **action potential** is generated.
8. The action potential travels rapidly along the axon, triggering vesicle release at its own downstream synapses.
9. The used vesicle is **retrieved and recycled**; new transmitter is synthesised and stored.

[Lecture slide omitted from the public copy.] The action potential travels along the axon.

The idea in plain words

Notice the conversion happening twice. An *electrical* signal arrives at the terminal, becomes a *chemical* signal to cross the cleft, and becomes *electrical* again in the next cell.
Step 9 is easy to skip past but explains why a synapse can keep firing: vesicles are recovered from the membrane and refilled rather than consumed once.

24. Excitation, Inhibition and Summation

[Lecture slide omitted from the public copy.] Postsynaptic stimulation changes excitability.

Two directions

Excitatory stimulation *depolarises* the membrane, making the action potential threshold *easier* to reach.
Inhibitory stimulation *hyperpolarises* it, making the threshold *harder* to reach.

[Lecture slide omitted from the public copy.] Three EPSPs summing to threshold.

The idea in plain words

An **excitatory postsynaptic potential** (EPSP) is the depolarisation produced by one excitatory input. A single EPSP does *not* usually reach threshold.
Neurons with many synapses require a **summation** of EPSPs — in the figure, three accumulate before the postsynaptic cell fires.

Watch out — summation is what makes a neuron a computing element

If one input reliably produced one output, a neuron would be a relay. Because threshold requires several inputs to add up, and because inhibitory inputs subtract, the cell is performing a weighted decision over everything arriving at once.
This is the biological fact underneath every model neuron that sums its weighted inputs and compares the total against a threshold.

25. What an Electrode Records

[Lecture slide omitted from the public copy.] Intracellular and extracellular recording, and the local field potential.

The idea in plain words

An electrode can be placed *inside* a cell or *outside* it, and the two give opposite-looking traces:

- **Inside** — positive ions flow in, so the recorded voltage *increases*.
- **Outside** — those same positive ions move *away* from the electrode, so the recorded voltage *decreases*.

Same event, mirrored signature, purely because of where the electrode sits.

Local field potential

Neurons are densely packed and highly interconnected, so an extracellular electrode *cannot tell which neuron* produced a voltage change. It measures the activity of a local neuronal environment — the **local field potential** (LFP).

Watch out — this limitation is the seed of the imaging problem

An extracellular electrode already cannot resolve single neurons. The lecture flags the LFP as "incredibly important when we talk about the basis of the MRI signal" — and fMRI is coarser still, summing over millions of cells and a slow vascular response.

The chain of losses is worth holding onto: single cell → local population → voxel. Each step trades spatial precision for coverage, and every interpretation of an imaging result inherits all of them.

Part VII

Neurotransmitters, Hormones and the Body

What crosses the cleft, how drugs interfere, and how the brain talks to the rest of the body.

26. Why Chemistry, Not Electricity

The idea in plain words

Neurons are separated by the synaptic cleft, and an electrical current cannot jump it. Chemical **neurotransmitters** carry the signal across.

The consequence is a division of labour worth stating plainly: the *electrical* mechanisms are highly similar from neuron to neuron, while the *chemical* ones vary enormously. Over **50 neurotransmitters** have been identified, each with slightly different neurochemistry and function.

So the diversity of brain signalling lives almost entirely in the chemistry.

[Lecture slide omitted from the public copy.] Categories of neurotransmitter.

Categories

Amino acids — glutamate, aspartate, glycine, D-serine.
Peptides — somatostatin, vasopressin, oxytocin, opioid peptides.
Others — serotonin, epinephrine, histamine, melatonin, and more.

[Lecture slide omitted from the public copy.] Excitatory, inhibitory, and both.

Watch out — the effect belongs to the receptor, not the transmitter

It is natural to read "excitatory neurotransmitter" as a property of the molecule. It is not. The effect depends on *which receptors the postsynaptic cell carries* — the same transmitter meets different receptors and does different things.
Dopamine is the lecture's example of a transmitter that is excitatory or inhibitory depending on the local receptor environment. The postsynaptic cell, by choosing its receptors, largely determines what a signal means to it.

27. Four Transmitters and Their Clinical Signatures

Acetylcholine

Excitatory. Activates the motor system; important for control of skeletal muscle and motor tone. Regulates attention, arousal, learning and memory.
Clinically: abnormally *low* levels are seen in **Alzheimer's disease**, alongside clear memory deficits.

[Lecture slide omitted from the public copy.] Dopamine.

Dopamine

Critical for motor control, especially fine movement. Also involved in reward and positive emotion.
Clinically: *low* in **Parkinson's disease**, particularly in the **substantia nigra** which produces it — explaining the parkinsonism and bradykinesia. *Elevated* in **schizophrenia**, particularly frontally.

Glutamate

The most common neurotransmitter in the nervous system; central to learning and memory.
Clinically: excessive glutamate is **toxic** to neurons and is implicated in diseases such as **ALS**.

The idea in plain words

The pattern across all four is that *either* direction is pathological. Too little dopamine gives Parkinson's, too much is associated with schizophrenia; glutamate is essential and, in excess, kills the cells that need it.
This is why the clinical target is a *balance* rather than a maximum, and it is the reason so many neurological and psychiatric drugs aim at receptors rather than at transmitter levels directly.

28. Glutamate, AMPA and NMDA

[Lecture slide omitted from the public copy.] The two receptors and what each admits.

The idea in plain words

Glutamate binds two different postsynaptic receptors, and they behave differently:

- **AMPA receptor** — glutamate opens the channel, admitting **sodium**.
- **NMDA receptor** — holds a **magnesium plug** that blocks ion flow. Glutamate binding expels the plug, admitting **calcium**.

The combined influx of sodium and calcium triggers downstream chemical changes that are critical for **long-term potentiation** (LTP), the mechanism underlying learning and memory.

Watch out — the magnesium plug is what makes NMDA a coincidence detector

The plug means the NMDA receptor needs *two* things at once — glutamate binding, and the cell already depolarised enough to dislodge the magnesium.

A receptor that only responds when input and postsynaptic activity coincide is a molecular implementation of "fire together, wire together". This is the cellular machinery behind Hebbian learning, and behind the LTP mentioned above.

29. How Drugs Interfere

[Lecture slide omitted from the public copy.] Unbound, agonist, receptor agonist, competitive and non-competitive antagonist.

Four ways to act on a receptor

Unbound — channel closed, nothing crosses.
Agonist — the natural transmitter binds and opens the channel.
Receptor agonist — a drug that mimics the transmitter, binding and opening the channel artificially. Useful where a patient is deficient in the natural transmitter.
Competitive antagonist — a drug that binds *the same site* but keeps the channel closed, blocking the natural transmitter from binding.
Non-competitive antagonist — binds a *different* site, and prevents the transmitter from having its effect without competing for its place.

The idea in plain words

This is the whole logic of neuropharmacology on one slide. If a system is underactive, supply an agonist; if overactive, block it with an antagonist.

The competitive/non-competitive distinction matters practically: a competitive antagonist can be out-competed by raising transmitter levels, while a non-competitive one cannot, because it is not fighting for the same site.

30. Talking to the Rest of the Body

The idea in plain words

Two routes out of the brain:

- **Direct innervation** — through the spinal cord and peripheral nervous system. Fast, targeted, point-to-point.
- **Hormones** — secreted into the blood and diffusing throughout the body. Slow, broadcast, and reaching everything the circulation reaches.

[Lecture slide omitted from the public copy.] The twelve cranial nerves and the corticospinal tracts.

Direct innervation

Twelve cranial nerves extend from the brainstem to the face and upper body.
Corticospinal tracts run down the spinal cord, carrying sensory and motor information between the body and the brain — reaching as far as the tailbone, and serving the gut, respiration, heart and lungs.

[Lecture slide omitted from the public copy.] The endocrine system.

Hormones

Signalling molecules produced by glands throughout the body — collectively the **endocrine system** — transported by the circulatory system to distant target organs.
They regulate heart rate, breathing, digestion, metabolism, sleep, reproduction and mood.

31. The Blood–Brain Barrier

The blood–brain barrier

Endothelial cells lining the blood vessels prevent bacteria and large molecules from entering the brain, protecting it from infection and toxins.
Only small hydrophobic molecules such as **oxygen** pass freely; others, such as **glucose**, are **actively transported** across.

Watch out — a protective barrier is also a delivery problem

The barrier keeps toxins out — and keeps most drugs out too. A compound that acts perfectly on neural tissue in a dish is useless if it cannot cross.
It also creates the puzzle the lecture raises next: if hormones circulate in the blood and the blood is walled off, how do hormones reach the brain at all?

The idea in plain words

The answer is that parts of the brain sit *outside* the barrier. The **hypothalamus**, **pituitary** and **pineal** glands are brain structures that are not protected by it, and they secrete hormones into the blood themselves.
Through them the brain regulates temperature, thirst, hunger, circadian rhythm and the stress response.

32. The Loop Back: Adrenaline

[Lecture slide omitted from the public copy.] From adrenal gland to memory.

The idea in plain words

Hormonal communication is **bidirectional** — the body signals back to the brain. The lecture's example is worth following in full, because it shows how the barrier is circumvented. In danger, the **adrenal glands** secrete **adrenaline** into the blood, activating the **sympathetic nervous system**: pupils dilate, saliva is inhibited, respiration and heart rate rise — the flight response.
But **epinephrine cannot cross the blood–brain barrier**. Instead it binds receptors on the **vagus nerve**, which releases **glutamate** onto synapses in the brainstem; projections from there reach memory areas.

Watch out — this is why frightening events are remembered vividly

The pathway ends in memory structures, and that is not incidental. An animal that survives a danger has a strong reason to encode it.
Note also the elegance of the workaround: a molecule locked out of the brain gets its message in anyway, by triggering a nerve on the outside that releases a permitted transmitter on the inside. The barrier is respected and the signal still arrives.

Part VIII

Functional Anatomy

What the tissue types are, and the systems whose function is well established.

33. Sources of Variety

The idea in plain words

Neurons differ along several independent dimensions, and each one contributes to functional specialisation:

- **Type** — sensory, inter-, motor.
- **Length and complexity** — from a few millimetres to *three feet* for spinal neurons reaching the leg muscles; from dense dendritic sprouting to only a few dendrites.
- **Receptors and transmitters** — which receptors a cell carries determines which transmitters it responds to.
- **Sign** — the receptor–transmitter combination determines whether the neuron excites or inhibits its network.

[Lecture slide omitted from the public copy.] Cortical organisation in columns and layers.

Columns and layers

Columns — where information processing takes place.
Layers — most often involved in transmitting information from one area of a network to another.

34. Three Tissue Types on a Scan

[Lecture slide omitted from the public copy.] Cerebrospinal fluid, dark on these images.

Cerebrospinal fluid (CSF)

Fluid surrounding the brain, in which it effectively floats. Produced by the **choroid plexus** within the **ventricles**.
Three roles: **protection** from external trauma — so the brain does not strike the inside of the skull when the head moves; supplying **nutrients**; and **waste disposal**.

Grey and white matter

Grey matter — neuronal *cell bodies*. The information *processing* component.
White matter — bundles of *axon projections*. The information *transmission* component.

The idea in plain words

Three tissue types, three appearances, and the distinction is functional rather than merely visual: CSF is dark, grey matter processes, white matter transmits.
This connects back to the myelin discussed earlier — white matter is white because glia have wrapped those axons in fat.
Named white-matter bundles include the **anterior** and **posterior cingulate**, and the **corpus callosum**, the major connection between the two hemispheres.

35. The Two Homunculi

[Lecture slide omitted from the public copy.] Primary motor cortex.

Primary motor cortex

The dorsal portion of the frontal lobe, critical for **planning and execution of movement**. Its neurons send axons down to the spinal cord to initiate and sustain muscle movement.
Arranged in an orderly map from head to toe: **foot, leg and hip most medial**; **hand, face and tongue more lateral**.

[Lecture slide omitted from the public copy.] Primary somatosensory cortex.

Primary somatosensory cortex

Immediately **posterior** to the primary motor cortex. Receives sensory input from the whole body in an equally orderly tactile map.

Watch out — the map is scaled by sensitivity, not by size

The amount of cortex given to a body part is *not* proportional to that part's physical size. It is proportional to the **density of tactile receptors** there.
So the face, nose, lips and eyes command far more cortex than the shoulder or torso — which is exactly why fingertips and lips discriminate fine detail while the back does not. The distorted homunculus figure is a map of resolution, not of anatomy.

36. The Thalamus, and Crossing Over

[Lecture slide omitted from the public copy.] The thalamus.

Thalamus

Sits at the centre, at the base of the brain; consists of a number of **subnuclei**. The brain's **relay station**, passing sensory and motor information between the spinal cord and the motor and somatosensory cortices. Also involved in regulating **sleep** and **consciousness**.

The idea in plain words

The left side of the brain receives sensory information from — and drives motor action in — the **right** side of the body, and vice versa. The pathways cross at the level of the **medulla**. This is why **contralateral** appears so often in results, and why a left-hemisphere stroke produces right-sided weakness.

37. Visual Cortex

[Lecture slide omitted from the public copy.] Visual cortex: columns, and organisation by visual field.

The idea in plain words

Highly organised in **columns**, processing orientation, motion and colour. Organised by **visual field**: the *right* visual field is processed by the *left* hemisphere, and vice versa — the same crossing seen for touch and movement.
Within a column, each neuron is sensitive to a particular **orientation**. No single cell represents the image; the *population* firing across many differently tuned neurons represents orientation, direction, colour and shape.

[Lecture slide omitted from the public copy.] The dorsal and ventral streams.

Two streams

Primary information is processed in **V1**, then divides:
Dorsal / "where" — V2, MT, MST and parietal regions. Direction, speed, where an object is and where it is going.
Ventral / "what" — V2, V4 and down to the medial temporal lobe. Object identity, recognition, complex shapes and body parts.

The idea in plain words

The same retinal input is split according to *what the information will be used for*: acting on an object versus identifying it. Location and identity are handled by anatomically separate pathways.

38. Cerebellum

[Lecture slide omitted from the public copy.] The cerebellum.

Cerebellum

A separate structure at the base of the brain with a complex organisation. Receives information from sensory systems and the spinal cord. Important for **posture**, **balance**, **coordination** and smooth movement.

Watch out — an honest "we don't fully know"

Recent work also implicates the cerebellum in **cognition** — but the lecture says plainly that exactly how it supports those functions "remains somewhat unclear".
Worth noting as a marker of where the settled anatomy of this module ends. The motor role is well established; the cognitive role is an active question.

Part IX

Cognitive Domains

What damaged brains revealed about language, personality and memory.

39. Cognition

Cognition

The mental action or process of acquiring knowledge and understanding through thought, experience and the senses — usually taken to include **reasoning**, **memory**, **attention** and **language**.

The idea in plain words

Almost everything in this part was learnt from patients with brain damage. That is worth registering before the case histories start: the method is to observe what a person can no longer do, and infer what the missing tissue was for.

40. Language: Broca and Wernicke

[Lecture slide omitted from the public copy.] Broca's patients and the area named after him.

The idea in plain words

Pierre Paul Broca, a French surgeon and anatomist, described two patients in 1861. **Leborgne** could produce no words or phrases — only the single syllable "tan", which became his nickname. **Lelong** retained a handful of simple words and verbs.
At autopsy both had damage to the same place: the **posterior inferior frontal gyrus**, now **Broca's area**.

Broca's aphasia

Patients *understand* words and simple sentences and know what they want to say, but cannot generate fluent speech or build the grammatical structure of a sentence.

[Lecture slide omitted from the public copy.] Wernicke's area.

The idea in plain words

Carl Wernicke, a German psychiatrist and neuropathologist, noticed patients with language problems but *no* damage to Broca's area. At autopsy the damage was in the **posterior superior temporal gyrus** — **Wernicke's area**.

Wernicke's aphasia

Also called **receptive aphasia**. Comprehension of spoken and written language is impaired. Patients *can* generate words and basic language, but cannot parse sentence structure to extract meaning.

[Lecture slide omitted from the public copy.] Receptive and expressive language as a network.

The idea in plain words

Combining the two findings gives a circuit rather than two isolated spots:

- **Receptive** — what was said or read: **Wernicke's area**, posterior.
- **What to say** — formulated in frontal areas.
- **Expressive** — how to say it: **Broca's area**.
- Output routes to **motor cortex** for speech or handwriting — or, for silent thought, back into the language areas.

Watch out — the two aphasias are near-mirror images

Broca: comprehension intact, production broken. Wernicke: production fluent, comprehension broken.
That double dissociation is what makes the pair convincing evidence for separable systems — far stronger than either case alone. If one area were simply "the language area", damage anywhere in it would impair language as a whole.

41. Personality: Phineas Gage

[Lecture slide omitted from the public copy.] The 1848 accident and the trajectory of the rod.

The idea in plain words

In 1848 **Phineas Gage**, a railroad worker, was tamping explosive into a hole when a spark ignited the powder and drove the iron rod through his skull — entering on the left side of his face and exiting through the top of his head, causing extensive damage to the **frontal lobes**. The rod was found some 80 feet away.
He survived. His *speech and motor control were largely preserved*, and he could understand others and express himself. But he underwent a severe **personality change**, becoming irritable and aggressive, and by his peers' account hard to get along with.

Watch out — what is preserved is as informative as what is lost

Gage matters because of the dissociation, not the drama. Language, movement and comprehension survived; personality and social conduct did not.
That is direct evidence that the frontal lobes support something separable from the abilities we most readily test — and it is why the assessment module treats **executive function** as its own dimension rather than folding it into general intelligence.

42. Memory: H.M. and E.P.

[Lecture slide omitted from the public copy.] Henry Molaison, and the extent of the resection.

The idea in plain words

Henry Molaison (H.M.) suffered severe epileptic seizures. In 1953 the surgeon **Scoville** removed, *bilaterally*, the **hippocampus**, **amygdala** and surrounding cortex — structures of the **medial temporal lobe**. The operation had been done before, but never on both sides at once.
Afterwards: language intact, normal IQ, normal working memory, motor control unaffected. But he could not learn or form **new** memories, and could not remember anything after the surgery or from a period before it. Only childhood memories remained.

The idea in plain words

In his own words: "Right now I'm wondering, have I done or said anything amiss? You see, at this very moment everything looks clear to me, but what happened just before? That's what worries me. It's like awakening from a dream. I just don't remember."

The idea in plain words

Patient E.P. independently confirms the finding. Herpes simplex encephalitis damaged his medial temporal lobes bilaterally — less extensively than H.M.'s surgery, and by an entirely different cause. Same profile: intact language, IQ, working memory and motor control; severe impairment in retaining new information.
Two different routes to the same lesion, producing the same deficit, is much stronger evidence than one remarkable case.

[Lecture slide omitted from the public copy.] The hierarchy of medial temporal lobe structures.

What the medial temporal lobe does

Critical for **forming new memories**, particularly memories for **facts and events** — episodes from one's personal life.
The **hippocampus** — Latin for sea-horse, after its shape — is the central structure.

43. The Mirror-Drawing Result

[Lecture slide omitted from the public copy.] Tracing a star between double lines, seen only in a mirror.

The idea in plain words

H.M. was asked to trace a star between two closely spaced lines without touching them — while seeing his hand only in a mirror. A hard task, made harder.
Over several days he made *fewer and fewer errors*. He was learning.
Asked whether he had done the task before, he said he had no memory of it.

Watch out — this single result reshaped the concept of memory

Performance improved while any recollection of practising was absent. The two came apart completely.
Brenda Milner concluded that there must be *multiple memory systems* — one for facts and events, dependent on the medial temporal lobe, and another supporting **motor learning**, which H.M. still had.
Before this, "memory" was treated as one faculty. Afterwards it could not be.

[Lecture slide omitted from the public copy.] Multiple memory systems.

Multiple memory systems

Medial temporal lobe — memory for facts and events.
Basal ganglia — including the **caudate nucleus**, **putamen** and thalamus. Critical for fine motor planning and movement.
Striatum — predominantly caudate and putamen. Involved in reward learning and reinforcement, and in motor and stimulus–response learning.

44. Highly Specialised Areas

[Lecture slide omitted from the public copy.] The fusiform face area and the extrastriate body area.

Two specialised regions

Fusiform face area — recognising faces.
Extrastriate body area — recognising bodies and body parts.

The idea in plain words

Damage to the fusiform face area causes **prosopagnosia**. Patients can see and describe the individual features of a face perfectly well, yet cannot recognise *whose* face it is — often

identifying people by voice instead.
The dissociation is precise: perceiving a face and identifying a face are separate operations, and only the second is lost.

Watch out — almost all of this came from damage

Language, personality, memory and face recognition were mapped from strokes, accidents, viruses and surgery. That is a powerful method — but it is opportunistic, the lesions respect no anatomical boundaries, and it can only study what happens to be destroyed.
Which is precisely the gap the next part addresses: neuropsychological assessment brings systematic measurement to the question, and neuroimaging eventually allows it to be asked of intact brains.

Part X

Neuropsychological Assessment

Measuring cognition systematically, and what has to be controlled for the measurement to mean anything.

45. What It Is For

Neuropsychology

An applied science concerned with the **behavioural expression of brain function** — in healthy people and, especially, in patients with impaired cognition.

The idea in plain words

Four purposes: **diagnosis** — inferring which disorder a patient has from behavioural performance; **patient care and planning**; **rehabilitation and treatment planning**; and **research** into which brain areas support which behaviour.

46. The Cognition–Performance Distinction

[Lecture slide omitted from the public copy.] Everything that stands between a cognitive ability and an observed behaviour.

Watch out — a poor score does not identify what is impaired

This is the central methodological point of the module. Suppose you test memory and the patient performs badly. Before concluding that memory is impaired, consider whether they:

- could **see and hear** the instructions
- could **attend** to the task
- could **perceive** the material
- were **motivated** to participate
- were **emotionally** able to follow instructions

- had the **motor capacity** to respond, if writing or drawing is required

Any of these can produce the same low score. You observe *performance*; you want to infer *cognition*; and everything on this list sits between the two.

The idea in plain words

Assessment therefore focuses on three dimensions of behaviour: **cognition**, **emotionality** and **executive function** — with emotionality included precisely because it shapes performance whether or not it is the thing being measured.

47. From IQ to Domains

The idea in plain words

Cognition was initially treated as a *single* ability — **intelligence**, measured as **IQ**. Standardised tests gave verbal and performance IQ on a scale centred at **100**: around 60–70 very low, 132–140 very high, with norms established from large population samples.
IQ remains a valid measure. It is not a sufficient one — it does not describe ability within specific domains.

[Lecture slide omitted from the public copy.] The specific cognitive domains now assessed.

Cognitive domains

Receptive functions — understanding spoken language and written text.
Memory and learning — retaining information over short and long intervals.
Thinking — problem solving, mental arithmetic, reasoning.
Expressive functions — producing a response, through motor control or expressive language.

The idea in plain words

The four map onto a sequence: take information *in*, *hold* it, *operate* on it, put a response *out*. A single summary score cannot distinguish a failure at one stage from a failure at another — which is the argument for domain-specific testing, and the same argument as the cognition–performance distinction, one level up.

48. What a Score Is Compared Against

The idea in plain words

Assessment **presupposes** some normal or prior level of function to compare against. Usually this is a **population average**: a test is administered to a large sample, a mean and standard deviation are established, and an individual is placed one, two or three standard deviations above or below it.
Norms are commonly stratified by **gender**, **age** and sometimes **education**, so the comparison group resembles the individual.

Watch out — comparison to a norm demands identical conditions

If a score is only meaningful relative to a population, then the testing conditions must match those under which the population was measured — lighting, a quiet room, and above all **strictly standardised instructions and administration**.
Deviate from the protocol and the comparison silently stops being valid. The number still looks like a number.

The idea in plain words

Better still is a person's *own* earlier score. **Longitudinal assessment** measures change within an individual — before and after a stroke or injury, say — and sidesteps the question of whether the population norm was ever the right comparison for them.
Such baselines are usually unavailable, which is why population norms do the work in practice.

49. Three Approaches

[Lecture slide omitted from the public copy.] Screening, hypothesis testing, and the battery.

Screening

Determines whether a person meets criteria — for study enrolment, or to decide whether further testing is warranted.
Flexible and efficient. Common instruments: **RBANS**, **MMSE** (Mini-Mental State Examination), **MoCA** (Montreal Cognitive Assessment).

Hypothesis testing

Assesses one specific question or domain in detail, using several tests to build a comprehensive picture of that single function.
Risk: it can overlook impairment in other domains — including domains that are contributing to performance on the one being measured.

Battery

Detailed testing across *all* cognitive domains, using a variety of tests per domain, giving a complete profile.
Cost: time-consuming, expensive, and tiring — sometimes six to eight hours.

The idea in plain words

The three trade breadth against cost, and the choice follows from the question. Screening asks *is anything wrong?*; hypothesis testing asks *how bad is this particular thing?*; the battery asks *what is the whole profile?*
Note that the hypothesis-testing risk is the cognition–performance problem again: testing memory alone cannot reveal that the real deficit was attention.

50. Memory Tests

[Lecture slide omitted from the public copy.] The Wechsler Memory Scale.

Wechsler Memory Scale

A comprehensive battery for learning and memory, with norms stratified by gender and age. Seven subtests:
Personal and current information — personal details, and facts such as the current president. **Orientation** — to time and place. **Mental control**. **Logical memory** — a paragraph-long story is read and the details recalled, repeated to see whether more is learnt with exposure. **Digit span** — recalling number sequences of increasing length. **Visual reproduction** — drawing simple figures from memory. **Paired associates** — learning word pairs, then being cued with the first word.

Word-list learning

A list of words is read and recalled; the list is presented repeatedly to measure learning across trials; after about 20 minutes, delayed recall is tested.
Gives measures of **learning**, **recall** and **recognition**, and separates impairments of **encoding** from those of **retrieval**.

Watch out — delayed recall is the sensitive measure

Several tests include a delay of around 20 minutes, and the lecture singles this out: **delayed recall** is a *particularly sensitive* measure of memory impairment.
The reason follows from H.M. Immediate repetition can be carried by working memory, which does not depend on the medial temporal lobe. Only after a delay does the task require what H.M. had lost — so the delay is what makes the test diagnostic rather than merely difficult.

51. Executive Function Tests

Executive function

Four components: **volition**, **planning**, **purposive action** and **effective performance**. Associated with the **frontal lobes**.

[Lecture slide omitted from the public copy.] The Stroop colour–word test.

The idea in plain words

The **Stroop** test has three conditions: read the colour *words* aloud; name the colour of ink that rows of Xs are printed in; and then name the ink colour of colour *words* printed in a conflicting colour.
The third is hard because reading is automatic. Performing it requires **inhibiting** the prepotent response — so the test measures **attentional and inhibitory control**, and does so by pitting two abilities against each other rather than by making the task merely effortful.

[Lecture slide omitted from the public copy.] The Tower of London.

The idea in plain words

Given a starting arrangement and a goal arrangement, the person determines how many individual moves are needed to get from one to the other. Difficulty scales by adding balls or requiring more moves.
This targets a different component — **planning**: working out a sequence before acting, rather than suppressing an unwanted response.

52. Where This Leaves Us

- **Neurons are separate, not continuous**, so signalling must convert electrical to chemical and back — which is where drugs get their purchase.
- **Summation makes a neuron a decision element**, since a single EPSP rarely reaches threshold.
- **An extracellular electrode records a population**, not a cell — the first of several losses of resolution on the way to a voxel.
- **The effect of a transmitter is set by the receptor**, which is why dopamine can excite or inhibit, and why both too much and too little of a transmitter is pathological.
- **The blood–brain barrier protects and obstructs**, and hormones reach the brain either through glands outside the barrier or by routes like the vagus nerve.
- **Cortical maps are scaled by receptor density**, not by body size.
- **Vision splits into "what" and "where"** streams according to the use the information will be put to.
- **Double dissociations carry the argument** — Broca against Wernicke, and H.M.'s intact motor learning against his lost episodic memory, which forced memory to be recognised as several systems.
- **Performance is not cognition.** Sensory, attentional, motivational and motor factors all sit between the ability and the score, and standardised conditions are what make a comparison to norms mean anything.

Part XI

Approaches to Neuroimaging

The available techniques, what each measures, and what each costs.

53. Structural and Functional

Two aims

Structural imaging — an image of the anatomy, and of pathology or injury.
Functional imaging — an image of the metabolic or pharmacologic response, or of cognitive function.

The idea in plain words

The techniques covered: **EEG**, **CT**, **SPECT**, **PET**, **MRI**, **fMRI**, **DTI** and **spectroscopy imaging**. The last four all run on the same MRI hardware.

54. EEG

[Lecture slide omitted from the public copy.] Electrode placement and a multi-channel readout.

The idea in plain words

Measures electrical activity through electrodes placed on the scalp, using **standardised positions** so that studies can be compared. Records at rest or in response to a stimulus, and assesses localised or network responses through temporal and spatial correlations.
Each line in the readout is one electrode. Widely used in **epilepsy** research to localise the focus of a seizure.

Watch out — EEG measures the thing the others only infer

Worth stating early, because the rest of this part drifts steadily away from it: EEG records *electrical activity* — the actual signalling. PET measures tracer uptake; fMRI measures blood.
What EEG gives up for that directness is spatial precision, since the signal has crossed skull and scalp before reaching the electrode. Directness and localisation trade against each other throughout.

55. CT

[Lecture slide omitted from the public copy.] X-rays through the volume from many angles, reconstructed into a 3-D image.

The idea in plain words

Computed axial tomography sends a series of X-rays through the body and measures how much each path absorbs. Repeating this from many angles allows a 3-D volume to be reconstructed from the cross-sections.
Fast and widely available — but it involves a **moderate dose of radiation**.

56. SPECT and PET

[Lecture slide omitted from the public copy.] SPECT: a gamma camera building 2-D images from multiple angles.

SPECT

Single photon emission computed tomography. A **nuclear imaging** technique: a gamma-emitting tracer is injected intravenously, either as a soluble ion or attached to a **ligand** that carries it. The scanner records gamma rays as a series of 2-D images from multiple angles.
Warm colours indicate greater uptake. **Spatial resolution is poor** — on the order of one to a few centimetres.

[Lecture slide omitted from the public copy.] PET: coincident gamma photons detected by a ring of cameras.

The idea in plain words

PET refines SPECT. The tracer emits a **positron**, which meets an electron and annihilates, sending out *two* gamma rays in opposite directions. A ring of gamma cameras detects the pair, and the tiny time difference between the two detections localises where along the line the annihilation happened.
Measuring a *pair* rather than a single photon is exactly why PET localises better than SPECT.

[Lecture slide omitted from the public copy.] Three tracers, three questions.

PET tracers

FDG-PET — glucose uptake. Relevant to tumours and to Alzheimer's disease, where reduced posterior uptake is visible early.
^{18}**F-AV-45** — binds **beta amyloid**, the misfolded protein of Alzheimer's disease.
^{11}**C-PBB3** — binds **tau**, a different misfolded protein.
The amyloid and tau markers are currently used predominantly for research.

The idea in plain words

The general principle is worth extracting: choose the biologically active molecule, and you choose what the image shows. PET is not one method but a family, and the tracer is the experiment.

57. MRI

Magnetic resonance imaging

Uses **static** and **variable** magnetic fields to perturb hydrogen atoms. The resulting magnetic resonance is measured by **radio-frequency coils** and converted into a 3-D image. Different **pulse sequences** generate different contrasts depending on tissue type.

[Lecture slide omitted from the public copy.] Inside the scanner: the always-on magnet, the gradient coils, and the radio-frequency coils.

The idea in plain words

The magnet is a wire loop — in some scanners up to **20 kilometres** of it — submerged in liquid helium at **−270 °C**, which makes the metal **superconducting** and sustains the field. It is *always on*. It is not switched on when a patient enters.

[Lecture slide omitted from the public copy.] A hospital bed pulled into the bore.

Watch out — the field is about 60,000 times the Earth's

Metallic objects nearby are extremely dangerous, which is why participants complete an extensive screening form. Patients with **pacemakers** must never undergo MRI — the field interferes with the device and the consequences can be life-threatening. Shunts and aneurysm clips also have to be checked.
The photograph of a bed suspended in the bore is both a safety lesson and the clearest available illustration of the field strength.

The idea in plain words

Given those precautions, MRI is remarkably benign: **no radiation**, and no known side effects even with repeated exposure. Good spatial resolution, and reasonable temporal resolution.
For brain work a **head coil** adds antennae close to the head, usually with a **mirror** so the participant can see out — which reduces claustrophobia, and doubles as the route by which stimuli are presented.

58. Three Refinements of the Same Hardware

[Lecture slide omitted from the public copy.] fMRI.

fMRI

Uses the differing magnetic resonance of **oxygenated versus deoxygenated blood** to image functional activity, on the reasoning that active tissue needs oxygenated blood.

[Lecture slide omitted from the public copy.] Diffusion tensor imaging.

DTI

Measures the **directionality of water diffusion**. Because white-matter fibre bundles are highly directional, this images **fibre projections** and the **integrity of white matter**.

[Lecture slide omitted from the public copy.] Spectroscopy imaging.

Spectroscopy imaging

Measures the magnetic signatures of **metabolites**, producing a frequency distribution for a defined location — the white box on the image. Used clinically in stroke and some cancers, and in research to relate metabolites to brain areas or functions.

The idea in plain words

fMRI, DTI and spectroscopy all use the *same hardware* as structural MRI. What differs is the pulse sequence and what is measured: blood oxygenation, water diffusion, or metabolite spectra.
That is why the following two parts spend so long on how the MRI signal is generated. Understand that one mechanism and four techniques come with it.

Part XII

The Basics of MRI

Where the signal comes from, why tissues differ, and how a measurement is given a location.

59. Spins in a Magnetic Field

[Lecture slide omitted from the public copy.] Protons precessing, parallel and anti-parallel.

The idea in plain words

Thermal energy makes protons spin. Placed in a magnetic field, each proton takes a state **parallel** (low energy) or **anti-parallel** (high energy) to the field direction, and precesses — spinning about its own axis while that axis itself sweeps around the field axis, conventionally called $\mathbf{B_0}$.
After a few seconds the system reaches equilibrium and all protons in the volume are aligned along the same direction.

Watch out — a static system carries no information

Once everything is aligned and steady, there is nothing to measure — and this is the pivot on which the whole method turns.
Change is required. Everything that follows exists to disturb the system and then watch it return.

[Lecture slide omitted from the public copy.] Excitation by a radio-frequency pulse.

Excitation

A small magnetic field applied **perpendicular** to B_0 — a **radio-frequency pulse** — pushes the spin out of alignment. A 90° pulse tips it fully into the transverse plane.
When the pulse stops, the spin precesses back to its original low-energy alignment.

60. Two Relaxations

[Lecture slide omitted from the public copy.] Longitudinal, or T1, relaxation.

[Lecture slide omitted from the public copy.] Transverse, or T2, relaxation.

T1 and T2

T1 — longitudinal relaxation. The component along the field axis. Immediately after a 90° pulse it is *zero*; it grows as the spin realigns, asymptoting to its maximum. The plotted signal *rises*.
T2 — transverse relaxation. The component in the transverse plane. Immediately after the pulse it is *largest*; it shrinks as the spin realigns. The plotted signal *falls*.

The idea in plain words

The two are the same motion resolved along two axes. As the spin tips back up, its vertical component grows while its horizontal component shrinks — which is why the T1 and T2 curves are mirror images.
A freely precessing magnetisation **induces a current** in a receiver coil, and that induced current is the measurement.

[Lecture slide omitted from the public copy.] Relaxation times by tissue and field strength.

The idea in plain words

Biological matter has **different but consistent** T1 and T2 relaxation times, depending on fat and water content — and the values vary with field strength, so the table is given for 1.5 T and 3 T separately.
Damadian's original 1979 paper turned on exactly this: **tumours relax differently from normal tissue**. In liver and in muscle, tumour values differ clearly from healthy values. The method was sensitive to tumour before it was used for anything else.

[Lecture slide omitted from the public copy.] White matter, grey matter and CSF have distinct relaxation properties.

[Lecture slide omitted from the public copy.] The two standard contrasts.

T1- and T2-weighted images

T1-weighted — CSF appears **black**; grey and white matter appear as shades of grey.
T2-weighted — CSF appears **bright white**; grey and white matter again as gradations of grey.

The idea in plain words

The logic of the whole imaging method sits here. Take a relaxation measurement at a location; compare it against known tissue values; conclude what tissue is there. Do that everywhere and an image assembles itself.
Nothing is being photographed. Every voxel is an *inference* from a timing measurement.

61. Giving the Measurement a Location

[Lecture slide omitted from the public copy.] The Larmor frequency.

Larmor frequency

Not any radio-frequency pulse will flip a spin. The efficient frequency is

$$f = \gamma B$$

— the **gyromagnetic ratio** γ, a known constant for a given nucleus, times the local **magnetic field strength** B, giving a frequency in megahertz.

The idea in plain words

The lecture's analogy: pushing a child on a swing. Energy transfers efficiently only if the pushes arrive at the right moment and rhythm. The Larmor frequency is that resonant rhythm.
Since γ is fixed by the nucleus — usually hydrogen in water — the only free variable is the field strength B.

[Lecture slide omitted from the public copy.] Slice selection.

Watch out — this is the trick that makes imaging possible

Because the Larmor frequency depends on the *local* field, adding a **linear gradient** — a small extra field that varies along one direction — makes the resonant frequency vary across the volume.
A pulse at one specific frequency then excites only the slice where the local field makes that frequency resonant. Everything else is left alone.
The slice can be moved either by changing the gradient or by changing the pulse frequency. Both are shown in the figure, and both work.

[Lecture slide omitted from the public copy.] Static field, gradients, pulse — and a voxel.

The three ingredients

A **static field** creates precession. **Linear gradients** in x, y and z create spatial specificity.
A **Larmor-frequency pulse** selectively excites a slice.
Selective excitation gives one dimension (usually z); distinguishing signals by gradient field is **frequency encoding**; the third, perpendicular gradient is **phase encoding**.
A **voxel** is a three-dimensional pixel — the in-plane resolution times the slice thickness.

[Lecture slide omitted from the public copy.] From the earliest scans to modern resolution.

The idea in plain words

Varying gradients, excitation and relaxation timing defines different **pulse sequences**, each sensitive to different tissue properties — T1-weighted, T2-weighted, proton density, and many more.
The comparison against one of the first MRI images is a fair measure of how far refinement of these sequences has carried the method.

Part XIII

The Basis of the fMRI Signal

BOLD, T2, the haemodynamic response, and what the signal is actually correlated with.*

62. Oxygenated and Deoxygenated Blood

[Lecture slide omitted from the public copy.] Rest, activation, and the influx that follows.

The idea in plain words

Three states in sequence:

1. **Rest** — a baseline ratio of oxygenated to deoxygenated haemoglobin.
2. **Active** — neurons consume oxygen, so locally there is *more* deoxygenated haemoglobin.
3. **Replenished** — an influx of oxygenated blood arrives, changing the ratio again.

[Lecture slide omitted from the public copy.] The BOLD signal.

BOLD

Blood Oxygen Level Dependent imaging. Oxygenated and deoxygenated haemoglobin differ in how much they **dephase** the signal — deoxygenated causes more — so the measurement is of **changes in the homogeneity of the magnetic field** within a volume. That measure is called **T2***.

[Lecture slide omitted from the public copy.] T2 against T2.*

T2 versus T2*

T2 — transverse magnetisation decay of a spin after the radio-frequency pulse.
T2* — transverse decay arising from *local magnetic field variations*: the field is not perfectly homogeneous, and is further distorted by local cellular and molecular conditions.

[Lecture slide omitted from the public copy.] Paramagnetic and diamagnetic haemoglobin.

The idea in plain words

Deoxygenated haemoglobin is **paramagnetic** — it distorts the local field, and that distortion is what T2* picks up.
Oxygenated haemoglobin is weakly **diamagnetic** and barely distorts the field at all.
So more deoxygenated blood means more inhomogeneity, more dephasing, and a measurably different signal.

63. The Haemodynamic Response Function

[Lecture slide omitted from the public copy.] The shape of the response.

The idea in plain words

Following the lecture's finger-tapping example through the curve:

1. The task begins; oxygen is removed from the blood, producing an **initial dip**.
2. The vascular system delivers an influx of oxygenated blood, and the signal **rises** to a peak.
3. The task stops, and the signal **falls**.
4. It **undershoots** below baseline for a few seconds before returning to rest.

This shape is the **haemodynamic response function** (HRF).

Watch out — the undershoot is unexplained

The lecture is candid that why the signal dips below baseline before recovering "is unclear". Worth registering: the HRF is used as a known, fixed template throughout fMRI analysis, and part of its shape is still not accounted for mechanistically.

[Lecture slide omitted from the public copy.] The central caveat.

Watch out — BOLD does not measure neural activity

Stated in the lecture and worth repeating whenever an fMRI result is interpreted: BOLD fMRI measures **metabolic demand** — oxygen consumption — not neural firing.

The HRF represents a *change in blood* triggered by neural activity. Every fMRI claim is therefore an inference across that gap.

64. Why Blood Follows Activity

[Lecture slide omitted from the public copy.] The physiological chain.

The idea in plain words

The mechanism runs through the **astrocytes**, connecting back to the synaptic biology of Part VI:

1. An action potential releases transmitter into the synaptic cleft.
2. Transmitter binds postsynaptic channels, admitting ions.
3. **Astrocytes** take glutamate back out of the cleft — which triggers glucose metabolism.
4. Astrocytes pump ions out to restore the local gradients.
5. All of that consumes **glucose and oxygen**, drawn from the blood.

So the oxygen demand that BOLD detects is generated largely by the housekeeping work of clearing up after signalling, rather than by the signalling itself.

65. What Is BOLD Correlated With?

The idea in plain words

BOLD was initially assumed to track **action potentials** — more spikes, more signal. **Logothetis** and colleagues (2001) tested this directly, recording BOLD alongside electrophysiology in monkey cortex, and comparing two measures:

- **Multi-unit activity** (MUA) — the firing of a number of neurons: *output.*
- **Local field potentials** (LFP) — a summation of postsynaptic potentials: *input and local processing.*

[Lecture slide omitted from the public copy.] BOLD against LFP and MUA.

[Lecture slide omitted from the public copy.] BOLD predicted from local field potentials.

The idea in plain words

BOLD correlated most closely with the **local field potential**, more so than with multi-unit activity. There is a temporal offset — the blood response is slower — but the correspondence is clear.
In the follow-up, a BOLD response *predicted* from the LFPs (grey) closely matches the BOLD actually observed (red).

Watch out — BOLD reflects input, not output

This is the most consequential result in the module. If BOLD tracks LFPs rather than spikes, it reflects the *input* to a neural population and the processing within it — rather than what that population sends onward.
An activated region may be receiving and processing heavily without increasing its own output. "This area lit up" does not mean "this area fired more".
And the lecture adds the necessary hedge: the correlations are *not perfect.* BOLD should never be treated as a direct measurement of LFPs either — it is a derived measure, most closely related to them, and representative of none of them exactly.

The idea in plain words

This also completes the chain of resolution losses begun in Part VI. A single cell gives way to a local population (the LFP), which gives way to a slow vascular response averaged over a voxel.
Each step buys coverage and costs precision, and an fMRI result carries all three losses at once.

Part XIV

Structural MRI Studies

Five ways to quantify anatomy, and the many reasons a brain can change size.

66. Qualitative Evaluation

[Lecture slide omitted from the public copy.] Cortical infarction, white matter hyperintensities, traumatic injury, periventricular lesions.

The idea in plain words

The simplest use of a structural scan is to look at it. Four examples:

- **Cortical infarction** — size and location assessed against comparable cases.
- **White matter hyperintensities** — areas of increased signal, counted and located.
- **Traumatic brain injury** — small areas of shearing in white and grey matter.
- **Periventricular white matter lesions** — common in ageing, appearing at the corners of the ventricles.

This can support a diagnosis — **Wilson's disease**, for instance, is characterised by hyperintensities visible on T1 or T2.

67. Volumetric Analysis

[Lecture slide omitted from the public copy.] Manual segmentation, and volumes compared between groups.

How a volume is computed

Delineate the structure of interest — manually, using a computer program — then **count the voxels** inside the outline and multiply by the volume of one voxel, which depends on in-plane resolution and slice thickness.

[Lecture slide omitted from the public copy.] Automated segmentation and labelling.

The idea in plain words

Automated methods separate white matter, grey matter and CSF, and can overlay anatomical labels across a large collection of structures — left and right separately — in an individual or across a group.
Different approaches suit T1- and T2-weighted scans, and a number of software packages will do it.

68. Cortical Thickness

[Lecture slide omitted from the public copy.] Pial and white matter surfaces, with vertices between them.

The idea in plain words

Find the boundary between white and grey matter, and the outer (pial) boundary of the brain, then measure the average distance between them along a stretch of cortex.
This isolates **grey matter** specifically — the cell bodies, the processing tissue — separating it from the white matter wiring.

[Lecture slide omitted from the public copy.] Cingulate thickness in super-agers, middle-aged and elderly controls.

The idea in plain words

Cortical **thinning** is a common outcome measure in neurodegenerative disease — Alzheimer's and Parkinson's especially.

69. Voxel-Based Morphometry

[Lecture slide omitted from the public copy.] Segmentation, normalisation, and whole-brain comparison.

Voxel-based morphometry

A whole-brain method asking *where* grey or white matter density differs between groups. Scans are segmented into tissue classes, **normalised** into a standard space, and compared against templates to locate differences.

Watch out — VBM answers a different question from volumetry

Volumetric and thickness analyses require you to *choose the structure first* — they quantify a region you already suspect.
VBM asks where in the brain the differences are, without a prior region of interest. Use it when you do not know where to look; use volumetry when you do.
The cost is the one every whole-brain method pays: testing everywhere means many comparisons, and the burden of controlling for them.

70. Morphology and Longitudinal Designs

[Lecture slide omitted from the public copy.] Shape differences, and shape change over 24 months in early Alzheimer's disease.

Watch out — shape can change while volume does not

This is the argument for morphometry over volumetry, and it is a good one. A structure can deflect *inward* in one place and *outward* in another, leaving total volume unchanged while the shape has altered substantially.
A volumetric analysis would report nothing. The longitudinal example — shape change in the entorhinal cortex across 0, 6, 12, 18 and 24-month scans in early Alzheimer's disease — shows a clear progressive pattern that a single volume number would have missed.

[Lecture slide omitted from the public copy.] Longitudinal structural studies.

The idea in plain words

Every method above can be run longitudinally: volume change in a structure of interest, cortical thickness over time, the proportions of tissue classes, **atrophy**, and shape.
As in the assessment module, measuring a person against their own earlier scan is stronger than measuring them against a population.

71. Relating Structure to Behaviour

[Lecture slide omitted from the public copy.] Three published examples.

The idea in plain words

Structural measures become interesting when placed against behaviour — task performance, pathological vulnerability, or intervention outcome. The lecture's examples:

- Brain structures **differ between musicians and non-musicians**.
- **Aerobic exercise training increases brain volume** in ageing humans.
- Abnormal cortical thickness and brain–behaviour correlations after **heavy prenatal alcohol exposure**.

The questions it opens: does cortical thickness change with antidepressant treatment? Does mid-life thinning predict later Alzheimer's disease? Do people with a larger hippocampus perform better on memory tasks?

72. Limitations

[Lecture slide omitted from the public copy.] Reasons a brain changes that may have nothing to do with your hypothesis.

Watch out — many things change brain volume

Development and atrophy are usually the intended signals. These are not:

- **Exercise** — shown above to *increase* volume.
- **Substance use** — both directions. Some prescribed antidepressants increase volume in certain areas; alcohol and drug abuse decrease it.
- **Inflammation or oedema** — fluid absorbed into tissue increases size overall. An infection elsewhere in the body can do this.
- **Gliosis** — disease affecting *glia* rather than neurons can change volume substantially with the neurons intact.

The idea in plain words

Gliosis is the subtlest of the four and the one most likely to be misread. Volume loss is naturally interpreted as neurons dying — but glia outnumber neurons, and a purely glial process produces the same volumetric signature.

A study concluding "neurodegeneration" from a volume difference may be measuring something else entirely. The others are covariates to record; gliosis is a mechanism to rule out.

Part XV

fMRI Studies and Experimental Design

From a voxel time series to a conclusion, and the assumptions that carry it.

73. What fMRI Buys

[Lecture slide omitted from the public copy.] The advantages.

Why BOLD fMRI is used

Non-invasive, unlike single-cell recording. Spatial resolution **1–1.5 mm**. Temporal resolution **~0.5 s or greater**. No known long-lasting effects, even across repeated scans — unlike X-ray or CT. Some report transient headache or dizziness at 7 T.
Critically, it images **whole-brain networks**, not only local activity.

The idea in plain words

The trade against single-cell recording is explicit: fMRI gives up directness and gains coverage, safety and repeatability. It sees the whole brain at once, which no invasive method can.

74. From Voxels to Time Series

[Lecture slide omitted from the public copy.] Image size, matrix size, and the resulting resolution.

The idea in plain words

A matrix is laid over the image. A **coarse** matrix gives large voxels and low resolution; a **fine** matrix gives more detail from the same image — and takes longer to acquire.
Resolution against acquisition time is the first design decision.

[Lecture slide omitted from the public copy.] One voxel, measured repeatedly.

TR and the time series

TR (time repetition) — the time the scanner takes to acquire a new image.
For one voxel, measuring T2* decay every TR produces a **time series**: that voxel's activation over the length of the scan.

The idea in plain words

This is done for *every* voxel in the volume. One scan therefore yields as many time series as there are voxels — which is the origin of the "big data" claim from the very first module.

75. The Block Design

[Lecture slide omitted from the public copy.] A finger-tapping experiment: tap, rest, tap, rest.

The idea in plain words

The simplest experiment. Ask a participant to alternate between tapping and resting, and look for voxels whose activation is *higher* during tapping than during rest.
Sustained tapping produces a train of haemodynamic responses which **concatenate** into a roughly box-shaped curve, offset in time because the blood response is slow. Find the voxels

whose time series shows that pattern — here, in the motor cortex — and you have localised the function.

Block design — advantages

Simple to design and implement. Analysis is straightforward: contrast on against off. **Robust activation**, because many events contribute. **Robust to uncertainty** in the timing and shape of the HRF, since it aggregates over many responses.

Block design — disadvantages

Must assume a **single mode of activation at constant level** — if the fifth tap differs from the first, the design fails. Cannot infer anything about **individual events** within a block. Cannot examine **variation in the HRF** itself.

The idea in plain words

Notice that the third advantage and the third disadvantage are the same property. Aggregating over responses makes the design insensitive to HRF variation — which is a strength if that variation is noise, and a weakness if it is what you wanted to study.

76. The Event-Related Design

[Lecture slide omitted from the public copy.] Task design convolved with the expected HRF gives the expected signal.

[Lecture slide omitted from the public copy.] Short events at variable intervals.

The idea in plain words

Events are short — typically a single trial, one finger tap — and spaced at **variable intervals**. The expected HRF is unchanged, but the spacing gives a different accumulated signal.
Individual events are recovered by defining an equation for each stimulus onset, multiplying by the expected HRF, and **deconvolving** the combined signal back into its constituent responses.

Event-related — advantages and disadvantages

For: inferences about **individual events** and about the **timing** of activation; flexible analysis including **post-hoc trial sorting**; random spacing avoids one trial confounding the next.
Against: **reduced power to detect** changes, since single events produce smaller responses; **sensitive to errors** in the assumed HRF; and **refractory effects** may carry from one trial to the next.

The idea in plain words

The two designs trade against each other cleanly. A block design asks *where*, with power to spare; an event-related design asks *what happened on this trial*, and pays for the privilege in sensitivity.
Post-hoc trial sorting is the distinctive gain: trials can be regrouped after the fact — remembered against forgotten, correct against error — which a block design forecloses at the moment it is designed.

77. Cognitive Subtraction

[Lecture slide omitted from the public copy.] Both designs rest on the same subtraction.

Watch out — there is no "off"

A voxel's time series shows activity *always*. There is no region that is silent until a task begins.
So fMRI never finds "the area that is active". It finds the *difference* between two conditions, and attributes that difference to whatever the conditions differed by. Every fMRI result is a subtraction.

Cognitive subtraction, and what it assumes

Also called a **categorical design**. Two assumptions:
Independence in space — different cognitive components occupy different places, so memory can be localised to the hippocampus, motor control to the motor strip, and the two dissociated.
Pure insertion (Donders, 1868) — processes in a complex condition are simply *added on top of* simpler ones. A button press accompanying an executive task is assumed to be the same button press as one accompanying arithmetic.

Watch out — pure insertion is an assumption, not a fact

It amounts to assuming cognition is **modular** — that complex tasks are combinations of simpler ones that can be added and removed without altering each other.
If a button press is executed differently while under executive load, the subtraction leaves a residue that gets attributed to the cognitive construct. The whole method rests on a 19th-century assumption that is convenient, productive, and not obviously true.

78. Choosing a Control Condition

[Lecture slide omitted from the public copy.] Why rest is a poor baseline.

Watch out — rest is not a control condition

Subtraction is valid only if the two conditions differ in *one* property. Anything that covaries with the manipulation is a confound.
Which makes "rest" a poor baseline, and the lecture cites work testing this explicitly. During rest people are encoding their surroundings and, above all, **mind-wandering** — rehearsing the task just finished, anticipating the next, remembering the shopping. Subtracting rest

removes a great deal that was never inactive.
The stronger design contrasts two *active* conditions and never uses rest.

[Lecture slide omitted from the public copy.] Faces against scrambled faces.

The idea in plain words

The model example. To find face-selective cortex, contrast **faces** against **scrambled faces** — the same pixels, the same contrast, the same luminance, rearranged into nothing. Every low-level visual property is matched, so the only remaining difference is whether the image is a face. What survives the subtraction is the **fusiform face area** met in Part IX. Extending it to faces, objects and scrambled versions of both allows face- and object-selective regions to be separated in one experiment.

79. Where This Leaves Us

- **The techniques trade directness against localisation.** EEG records electrical activity but localises poorly; PET and fMRI localise well but measure tracers and blood.
- **fMRI, DTI and spectroscopy share MRI's hardware**, differing in pulse sequence and in what is measured.
- **An MRI image is an inference**, not a photograph — every voxel is a tissue type deduced from a relaxation time.
- **Gradients make imaging possible**, because they make the resonant frequency vary with position, so a single frequency excites a single slice.
- **BOLD measures metabolic demand**, and correlates best with *local field potentials* — input and local processing — rather than with spiking output.
- **Structural measures come in five flavours**, and morphometry exists because shape can change while volume does not.
- **Many things change brain volume**, and gliosis in particular can mimic neurodegeneration.
- **Block designs ask where; event-related designs ask what happened on this trial**, and the same aggregation that gives one its robustness denies it the other's precision.
- **Every fMRI result is a subtraction**, resting on pure insertion — which is why the control condition, not the task, usually decides what a study can conclude.

Part XVI

Designing an fMRI Experiment

Technical limits, the participant's actual mental state, and five statistical designs.

80. The Goal, and Three Kinds of Constraint

What an fMRI experiment is trying to do

Use stimuli to **induce a psychological state**, then use the MR signal to measure the brain activity associated with it — so as to conclude where and how that process is supported.

[Lecture slide omitted from the public copy.] Three categories of consideration.

The idea in plain words

The considerations divide into **technical limitations**, **psychological state**, and **statistical design**. The first is about the scanner, the second about the person inside it, and the third about the analysis that will follow.

81. Technical Limitations

[Lecture slide omitted from the public copy.] The constrained environment.

The idea in plain words

The participant lies in a narrow bore. Stimuli are shown by a projector *outside* the scanner room — so as not to interfere with the magnetic field — onto a screen viewed through a mirror on the head coil. Responses come from button boxes with typically **three or four buttons** per hand.
Two design consequences follow directly: visual stimuli must be **large** enough to read through a mirror, and the number of **response options** is capped at three or four.

Timing, runs and length

Stimulus spacing — in an event-related design the interval between stimuli must be *varied*. A fixed interval leaves too little variability in the signal to deconvolve individual haemodynamic responses.
Run length — typically **4–6 minutes**, because of **scanner drift**: continuous operation heats the scanner, which changes the field, so stimuli at the end of a long run are not measured under the same conditions as those at the start. Runs are stopped to let it normalise.
Total length — around **60 minutes** or less, given how uncomfortable and noisy the environment is.

Watch out — motion is measured in the same units as the signal

Voxels are 1–1.5 mm. Participant movement is routinely *several* millimetres — so a shift can exceed the size of the thing being measured. Rotation does the same, and motion blurs structural scans visibly.
Padding around the head reduces it, at the cost of comfort — which in turn argues for shorter experiments. The constraints interact: a longer session gives more data *and* more motion.

82. Is the Participant in the State You Think?

[Lecture slide omitted from the public copy.] A three-second trial, and a decision made in 500 ms.

Watch out — the psychological state may be shorter than the trial

The lecture's example is worth following exactly. A participant is shown a Rubik's cube and asked whether they have seen it before in the experiment. Trial length: three seconds. They retrieve the answer and press the button in **500 ms**.
What happens in the remaining **2,500 ms**? Very likely: remembering being given a Rubik's cube aged five, wondering where it is now, whether it could still be found.
The scanner records all three seconds. Most of what it records is not the state under study.

[Lecture slide omitted from the public copy.] A Go/No-Go task, and what a predictable sequence turns it into.

Watch out — a shortcut changes what is being measured

In Go/No-Go, participants press for every letter except **X**, which measures **response inhibition**.
But if X appears as every fourth stimulus, or always after S, the sequence becomes *predictable* — and the participant is no longer inhibiting a response, they are **anticipating**. The task now measures prediction, and the paper will still say inhibition.
Stimulus ordering is not a housekeeping detail; it can silently redefine the construct.

[Lecture slide omitted from the public copy.] The full set of questions.

The idea in plain words

Two further considerations:
Complexity — a complex task takes longer to resolve than a simple one, so comparing them compares durations as well as processes. This is acute for **emotional** states, which are slow to induce *and* slow to switch: an angry state cannot be replaced by a happy one in the next trial without time in between.
Unintended states — a participant who cannot answer, or feels they are performing badly, may become frustrated or anxious about being judged. At that point the scanner is measuring frustration rather than the cognitive process.

83. Five Statistical Designs

[Lecture slide omitted from the public copy.] The common designs, and the trade-off underneath all of them.

Watch out — power against specificity

The trade-off runs through every design: **fewer conditions** give better **power** but less specific conclusions and less generalisability; **more conditions or levels** give worse power but better inference across stimulus types.
There is no design that escapes this, only designs that choose a point on it.

[Lecture slide omitted from the public copy.] Subtraction design: famous faces against unknown faces.

Subtraction design

Subtract activation during a control condition from activation during the experimental condition. Works best where the cognitive process is **categorical** — as in recognising *familiar* versus *unknown* faces.

[Lecture slide omitted from the public copy.] Factorial design: curvilinear against rectangular, open against enclosed, high against low ceilings.

Factorial design

Multiple levels of multiple factors, allowing the **interaction** between conditions to be tested.
Downside: interaction results can be very difficult to interpret — but if the interaction is the question, this is the only way to ask it.

[Lecture slide omitted from the public copy.] Parametric design: the N-back task.

The idea in plain words

In a **parametric design**, activity varies with difficulty or cognitive demand. The **N-back** task is the standard example: press the number you see (0-back), the number from the previous slide (1-back), or from two slides ago (2-back), with working-memory load rising at each step.

Watch out — psychological states may not be linear

The design assumes activation scales with load. At 4- or 5-back, participants simply *cannot* do the task — so activity would be *absent* rather than higher.
There is a point of optimal arousal beyond which performance and activation both fall away. A parametric design that runs past it produces a curve that looks like evidence and is an artefact of the ceiling.

[Lecture slide omitted from the public copy.] Individual differences: activation against behavioural performance.

Individual differences design

Correlate each subject's behavioural performance with their activation. Each point is one participant — in the example, caudate and hippocampal activation during spatial navigation plotted against recall performance.

[Lecture slide omitted from the public copy.] Outcome measures: scanning before and after an intervention.

Outcome measures design

A **within-subject** design: scan once before an intervention and once after, and use the difference to assess its effect.
Interventions used this way include drug development, practice or training, exercise, **deep**

brain stimulation (Parkinson's disease), **transcranial magnetic stimulation**, and **electroconvulsive therapy** (depression).

[Lecture slide omitted from the public copy.] Block, event-related, or hybrid.

The idea in plain words

The N-back is necessarily a **block** design — an instruction is given, then a run of stimuli share it. Questions about individual stimuli need an **event-related** design.
A **hybrid** pseudo-randomises events within blocks, so the same data can be analysed either way depending on the question.

The idea in plain words

The lecture's closing point is worth keeping: careful experimental design, and above all the *choice of control condition*, is critical to what a study can conclude — and getting it right takes research, practice and pilot testing before anyone enters the scanner.

Part XVII

Functional Connectivity and the Default Mode Network

What the brain does when you ask it to do nothing.

84. Correlation Instead of Contrast

[Lecture slide omitted from the public copy.] Functional connectivity MRI.

Functional connectivity MRI (fcMRI)

A variant of fMRI resting on the premise that brain activation is present **under any condition**, including the complete absence of a task.
Examining the brain at rest assesses its **functional organisation** independently of cognitive or behavioural performance.

The idea in plain words

The logic is **Hebbian**, and the lecture says so explicitly: cells that fire together wire together. Applied to voxels, two regions whose time series are **correlated** are taken to be functionally connected — contributing to the same task, or belonging to the same network.
This is the principle from the Hebbian learning material, repurposed as an analysis method rather than a learning rule.

Watch out — this inverts the whole method of the previous part

Task-based fMRI *contrasts* conditions and needs an experimental manipulation, a control condition, and the pure-insertion assumption.
Connectivity analysis *correlates* time series and needs none of them. It discards the subtraction logic entirely — which is what makes it usable in participants who cannot perform a task at all.
What it gives up is any direct claim about function. A correlation says two regions covary; it does not say what either is doing.

[Lecture slide omitted from the public copy.] Spatial and temporal correlation.

The idea in plain words

Correlations may be **spatial** — time series that rise and fall together — or **temporal**, offset slightly in time and still considered functionally connected.

85. Two Ways to Ask

[Lecture slide omitted from the public copy.] Voxel-to-voxel connectivity.

Voxel-to-voxel

Pick a single voxel as the **source**, then correlate every other voxel's time series against it. The resulting map colours each voxel by its degree of correlation with the source.

[Lecture slide omitted from the public copy.] Seed-based connectivity.

Seed-based

Select a **region of interest** — anatomically, or from task performance — usually by overlaying a sphere. Average the time series across the voxels in that seed, then look for regions correlating with that average.

[Lecture slide omitted from the public copy.] Left motor cortex against right motor cortex, and against left visual cortex.

The idea in plain words

The validation case. Left and right motor cortex show *high* correlation; left motor against left visual cortex shows *low* correlation.
That is the result you would want before trusting the method on anything unknown: homologous regions covary, unrelated ones do not. Seeds can equally be defined anatomically, by segmenting the structural scan and overlaying an atlas.

[Lecture slide omitted from the public copy.] Independent component analysis.

Independent component analysis

Rather than choosing a seed, ask how many **independent networks** the data contain — finding the maximum number of orthogonal components.
Commonly recovered: higher-order visual processing, lower-order visual processing, motor control, a **vigilance** (attention) network, error monitoring and response inhibition, and visual monitoring.

The idea in plain words

Note the difference in stance. Seed-based analysis tests a hypothesis you brought; ICA asks the data what structure it contains. It is the same distinction as volumetry against voxel-based morphometry, one level up.

86. The Default Mode Network

[Lecture slide omitted from the public copy.] The default mode network.

Default mode network (DMN)

A network of regions with highly correlated activation when a person is *not* engaged in a task. Typically involves the **posterior cingulate cortex** — often considered the network's hub — the **medial prefrontal cortex**, and temporal lobe regions.
Its defining property: it is **less active during an externally directed task** and more active at rest.

[Lecture slide omitted from the public copy.] Development, and cross-species overlap.

The idea in plain words

Two lines of evidence that the DMN is a real organisational feature rather than an artefact: **Developmental** — the correlated activity is not seen in very young children. It emerges roughly between ages **9 and 12** and persists through adulthood.
Cross-species — rat and monkey resting networks overlap substantially with the human DMN, in the same regions.

[Lecture slide omitted from the public copy.] Functions attributed to the DMN.

The idea in plain words

Unusually among functional networks, the DMN also overlaps considerably with **structural** connectivity — the correlated regions are physically wired together.
Functions attributed to it: comprehension, learning and memory; the **neurological basis of self** — self-reference and autobiographical information; thinking about the **past and future**; and **theory of mind**, social cognition and emotion.

Watch out — this is the mind-wandering that ruined rest as a baseline

Part XV said rest is a poor control condition because people mind-wander, rehearse and remember. The DMN is the network doing that — self-reference, autobiographical memory, imagining the future.
So the same activity appears twice in this course with opposite signs: as *contamination* to be designed around in task fMRI, and as the *object of study* in connectivity work. Whether resting activity is noise or signal depends entirely on the question.

[Lecture slide omitted from the public copy.] DMN deactivation and task performance.

The idea in plain words

Three findings link DMN *deactivation* to behaviour:

- Deactivation correlates with **successful memory encoding** — the network must switch off for an external stimulus to be taken in.
- Deactivation scales with **task difficulty** — more demanding tasks suppress it more.
- *Re*activation after learning is associated with better **retention and later recall**.

87. What Connectivity Studies Are Used For

[Lecture slide omitted from the public copy.] Superior temporal cortex connectivity in Alzheimer's disease, frontotemporal dementia and controls.

The idea in plain words

Resting data are collected by asking the participant to **stare at a fixation cross** — not falling asleep, not thinking about anything specific. The resulting time series are then correlated.
Four uses: relate connectivity to **brain structures** or to clusters from a task; assess it against **task performance**; and compare it **between groups**.

[Lecture slide omitted from the public copy.] DMN deactivation in young and older adults, against memory performance.

The idea in plain words

Younger adults show *greater* DMN deactivation, associated with better memory performance; older adults show *less*, associated with worse performance.
This combines the two uses at once — a group comparison and a correlation with behaviour measured outside the scanner.

[Lecture slide omitted from the public copy.] Where connectivity changes have been reported.

Watch out — a very long list is not automatically good news

DMN and connectivity changes have been reported in Alzheimer's disease, autism, depression, schizophrenia, ageing, epilepsy, Parkinson's disease, obsessive-compulsive disorder and anorexia nervosa.

A measure that differs in almost every disorder studied is sensitive but not very *specific* — it flags that something is different without identifying what. That is genuinely useful as a marker, and weak as a diagnosis.

88. Where This Leaves Us

- **The scanner constrains the experiment** — large stimuli, three or four response buttons, runs of 4–6 minutes against scanner drift, sessions under an hour, and motion measured in the same millimetres as the signal.
- **The psychological state may not fill the trial**, may be reached by a shortcut, may take longer for complex or emotional material, and may be replaced by frustration.
- **Every design trades power against specificity**, and the five common designs are five choices along that trade.
- **Parametric designs assume linearity** that fails once a task becomes too hard to do at all.
- **Connectivity analysis correlates rather than contrasts**, which frees it from the control-condition problem and costs it any direct claim about function.
- **The default mode network is what the brain does at rest**, is developmentally established, overlaps across species, and must deactivate for external information to be encoded.
- **Resting activity is noise or signal depending on the question** — the same mind-wandering that disqualifies rest as a baseline is the object of study in connectivity work.

> *To add from lectures: Next: structural connectivity studies based on diffusion tensor imaging.*

Figures are frames from the course slides (Arnold Bakker, Johns Hopkins University School of Medicine); the anatomical images they contain carry their own citations on the original slides.